Proving Basic Algorithms Correct: Sorting

Computer-Assisted Theorem Proving (7CCSACTP)

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Merge Sort: Top-Down

```
merge :: 'a list ⇒ 'a list ⇒ 'a list

merge [] ys = ys
merge xs [] = xs
merge (x # xs) (y # ys) =
  (if x ≤ y then x # merge xs (y # ys)
   else y # merge (x # xs) ys)

msort :: 'a list ⇒ 'a list

msort xs =
  (let n = length xs
   in if n ≤ 1 then xs
     else merge (msort (take (n div 2) xs))
                (msort (drop (n div 2) xs)))
```